

Photosynthesis — Lecture Notes

Class XI Biology · Unit 3 · Slide handout

1. What is Photosynthesis?

Photosynthesis is the process by which green plants, algae, and some bacteria convert light energy into chemical energy stored in glucose. It takes place mainly in the leaves, inside organelles called chloroplasts. The overall word equation is: carbon dioxide + water → glucose + oxygen, in the presence of sunlight and chlorophyll.

2. Where It Happens

The chloroplast contains a green pigment called chlorophyll, which absorbs light most strongly in the blue and red regions of the spectrum and reflects green light. Photosynthesis occurs in two main stages: the light-dependent reactions and the light-independent reactions (the Calvin cycle).

Figure 1: Overview of Photosynthesis

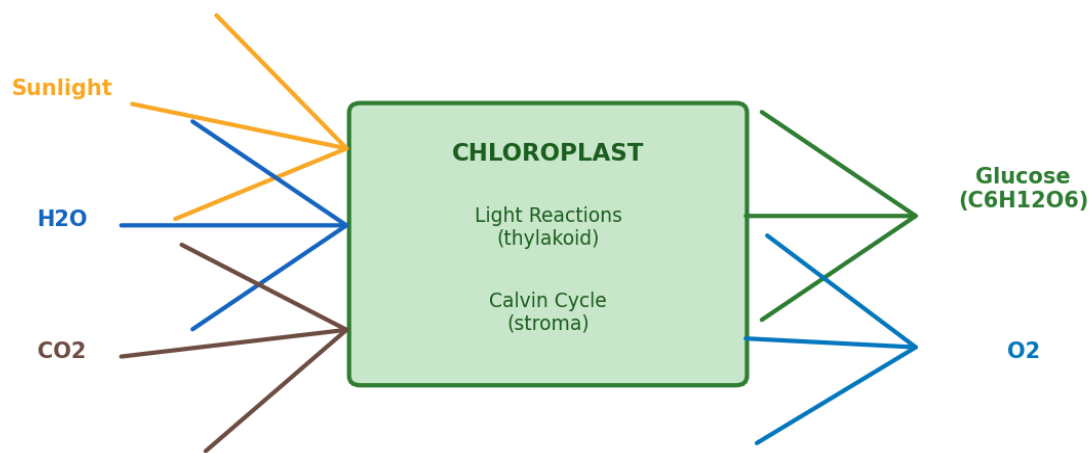


Figure 1: Inputs (sunlight, water, carbon dioxide) and outputs (glucose, oxygen) of photosynthesis.

3. The Two Stages (detail)

Light-dependent reactions: occur in the thylakoid membranes. Light energy splits water molecules (photolysis), releasing oxygen as a by-product and producing ATP and NADPH.

Light-independent reactions (Calvin cycle): occur in the stroma. ATP and NADPH from the first stage are used to fix carbon dioxide into glucose. This stage does not directly require light.

4. Factors Affecting the Rate

Three main factors affect the rate of photosynthesis: light intensity, carbon dioxide concentration, and temperature. Each can act as a limiting factor when in short supply.

Summary

Photosynthesis sustains nearly all life by producing oxygen and forming the base of most food chains.